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Antioxidant and neuroprotective effects of Dictyophora indusiata polysaccharide in Caenorhabditis elegans

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Abstract

Ethnopharmacological relevance

Dictyophora indusiata is a medicinal mushroom traditionally used in China for a variety of conditions, including inflammatory and neural diseases. D. indusiata polysaccharides (DiPS) are shown to have in vitro antioxidant activity but in vivo evidence is lacking. This study aimes to explore the antioxidant capacity and related neuroptotective activities of DiPS using wild-type and neurodegenerative Caenorhabditis elegns models.

Materials and methods

The antioxidant capacities of DiPS were first determined using paraquat survival and Pgst-4::GFP expression assays in wild-type and transgenic C. elegans models, respectively, and then further investigated by determining reactive oxygen species (ROS) level, malondialdehyde (MDA) content and superoxide dismutase (SOD) activity as well as functional parameters of mitochondria. The activation of stress response transcription factors and neuroptotective activities were examined using nuclear localization and chemosensory behavioral assays in transgenic nematodes, respectively.

Results

DiPS was shown not only to increase survival rate and reduce stress level under paraquat-induced oxidative conditions but also to decrease ROS and MDA levels and increase SOD activity in C. elegans models. Moreover, DiPS was also able to restore the functional parameters of mitochondria, including membrane potential and ATP content, in paraquat-stressed nematodes. In addition, nuclear translocation assays demonstrate that the stress response transcription factor DAF-16/FOXO was involved in the antioxidant activity of the

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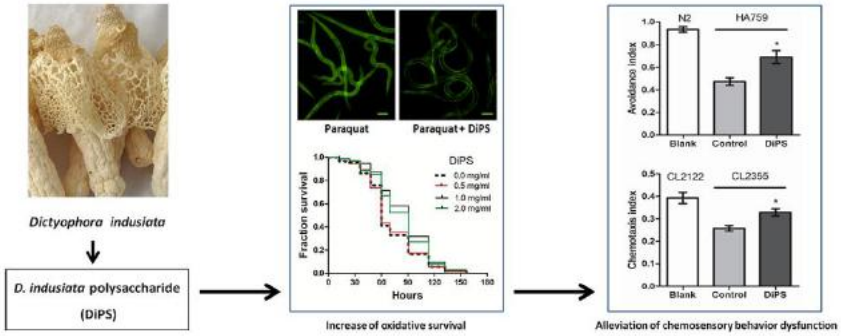
polysaccharide. Further experiments reveal that DiPS was capable of reducing ROS levels and alleviating chemosensory behavior dysfunction in transgenic nematode models of neurodegenerative diseases mediated by polyglutamine and amyloid-β protein.

Conclusions

These findings demonstrate the antioxidant and neuroprotective activities of the D. indusiata polysaccharide DiPS in wild-type and neurodegenerative C. elegans models, and thus provide an important pharmacological basis for the therapeutic potential of D. indusiata in neurodegeneration.

Graphical Abstract



Abbreviations: DiPS, Dictyophora indusiata polysaccharide; ROS, reactive oxygen species; MDA, malonaldehyde; SOD, superoxide dismutase; AD, Alzheimer's disease; PD, Parkinson's disease; HD, Huntington's disease; Aβ, amyloid-β protein; polyQ, polyglutamine.

Keywords: Dictyophora indusiata; oxidative stress; mitochondria; DAF-16/FOXO; neurodegenerative diseases

1. Introduction

Oxidative stress has been implicated in the pathogenesis of various disorders such as neurodegeneration (Xie et al., 2013) and cancer (Raina et al., 2013). Age-related

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neurodegenerative diseases, including Alzheimer's disease (AD), Parkinson's disease (PD) and Huntington's disease (HD), have become more prevalent as the population ages. Many neurodegenerative diseases are characterized by abnormal aggregation of pathogenic proteins such as amyloid-β protein (Aβ) in AD and polyglutamine (polyQ) in HD (Bourdenx et al., 2015; Chen et al., 2002). The protein aggregates are likely to stimulate reactive oxygen species (ROS) generation and lead to a cascade of oxidative damages to neurons (Facecchia et al., 2011). Interestingly, pharmacological intake or dietary supplementation of antioxidants may represent a therapeutic strategy for neurodegenerative diseases as a number of antioxidants have been shown to inhibit protein aggregation in addition to their antioxidant activity per se (Smid et al., 2012).

Recent studies suggest that polysaccharides from traditional medicinal plants have great potentials in the prevention of oxidative damages in diverse diseases. For example, the polysaccharides prepared from Angelica sinensis, a traditional herb used to tonify the blood, are shown not only to protect PC12 neuronal cells from hydrogen peroxide-induced cytotoxicity but also to prevent oxidative damages in cortical tissue of rats with focal cerebral ischemia (Lei et al., 2014). We have previously reported that astragalan, a polysaccharide isolated from the roots of adaptogenic Chinese herb Astragalus membranaceus, is able to alleviate the neurotoxicity mediated by polyQ aggregation in C. elegans models (Zhang et al., 2012). Interestingly, A. membranaceus polysaccharide is also shown to have antioxidant and anti-inflammatory activities (Huang et al., 2013).

Dictyophora indusiata (Vent. ex Pers) Fisch (Chinese name Zhu Sun, meaning the bamboo mushroom), synonymously called Phallus indusiatus, has a cosmopolitan distribution, including Africa, South America and South Asia, and has been consumed in a variety of folklore situations by ethnic groups (Ker et al., 2011; Kinge et al., 2011). In China, D. indusiata is regarded as the queen of mushrooms and has been traditionally used since

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Tang Dynasty as a medicine for a variety of conditions, including inflammatory, stomach and neural diseases (Hara et al., 1982; Ker et al., 2011; Lee et al., 2002). Modern studies suggest that a number of pharmacological functions of D. indusiata are associated with its polysaccharide fractions. For instance, D. indusiata polysaccharide is recently shown to have immunomodulatory activity through promoting macrophage NO, TNF-α, and IL-6 secretion (Liao et al., 2015). The polysaccharides isolated from D. indusiata are also reported to have antioxidant capacity in vitro (Deng et al., 2012; Ker et al., 2011), but their antioxidant activities in cellular and animal models are lacking. Here we first tested the capacity of D. indusiata polysaccharide (DiPS) against paraquat-induced oxidative toxicity in Caenorhabditis elegans, which is a major model organism used for a wide variety of physiological processes due to its advantages of small body size, short lifespan, ease of maintenance, and physiological similarity to mammals (Wang et al., 2014). Then we investigated the effect of DiPS on ROS level, malonaldehyde (MDA) content and superoxide dismutase (SOD) activity as well as on functional parameters of mitochondria, including mitochondrial membrane potential and ATP content. We also examined the activation of SKN-1/Nrf2 and DAF-16/FOXO, the transcription factors related to stress resistance and lifespan regulation. To test whether DiPS was able to inhibit polyQ- and Aβ-mediated neurotoxicity, we investigated the effect of the polysaccharide on chemosensory behavior dysfunction in transgenic HD- and AD-like C. elegans models.

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2. Materials and methods

2.1. Preparation of polysaccharides

The dry fruit body of Dictyophora indusiata (Vent. ex Pers) Fisch was purchased from Tongrentang Group Co., Ltd. (Wuhan, Hubei, China) and identified by Professor Keli Chen from Hubei University of Chinese Medicine. A voucher specimen (No. 28-DI-2012ZJ) has

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been deposited at the School of Pharmaceutical Sciences, Wuhan University. The polysaccharides were prepared essentially as described previously (Li et al., 2011). Briefly, 1.0 kg of fruit body was extracted three times in 5 L of 95% ethanol at 50°C for 4 h. The materials were then collected by filtration, air-dried and extracted four times with 5 volumes of H_2O (v/m) at 100°C. The supernatant was obtained by centrifugation (5000 g, 10 min), concentrated under reduced pressure at 40°C, and then precipitated with 4 volumes of ethanol (v/v). After removal of proteins using Sevag method, the polysaccharides were further purified using Amberlite 732 cation-exchange resin (Aladdin, Shanghai, China) column eluted with H_2O. The polysaccharides were then separated by anion-exchange chromatography on a DEAE-Sepharose Fast Flow (GE Healthcare, Uppsala, Sweden) column eluted with H_2O followed by 1.0 and 2.0 M NaCl sequentially. Neutral and acidic polysaccharides were obtained from H_2O and 1.0 M NaCl eluates, respectively, with a ratio of 1:3 (w/w), and the acidic polysaccharide fraction obtained was used in this study as D. indusiata polysaccharide (DiPS).

2.2. Analysis of sugar content and monosaccharide composition

The total sugar content of DiPS was determined by the phenol-sulphuric acid method using glucose as a reference at 490 nm (Huang et al., 1998). The monosaccharide composition of the polysaccharides was analyzed by gas chromatography as described previously (Huang et al., 1998). Briefly, the polysaccharides were first methanolyzed with 1 M HCl in anhydrous methanol at 80°C for 24 h with inositol as an internal standard. The mixture of methyl glycoside was then subjected to trimethylsilylation followed by analysis on a CP-3800 gas chromatography (Varian, Palo Alto, CA, USA) with a DB column (30 × 0.25 mm; Agilent, Palo Alto, CA, USA) (temperature programming: 140°C to 170°C at

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1.5°C/min, then to 250°C at 6°C/min, and then to 300°C at 30°C/min; nitrogen flow rate: 2.8 mL/min; ion source temperature: 300°C; injection temperature: 260°C).

2.3. Strains and maintenance

The following Caenorhabditis elegans strains were used in this study: wide-type N2, CL2166 {dvIs19[(pAF15)gst-4p::GFP::NLS]}, LG345 {geIs9[gpa-4p::skn-1b::GFP+rol-6(su1006)]}, GR1352 {xrIs87[daf-16(alpha)::GFP::daf-16B+rol-6(su1006)]}, HA759 {rtIs11[osm-10p::GFP+osm-10p::HtnQ150+Dpy-20(+)]}, CL2355 {dvIs50[pCL45(snb-1::Abeta1-42::3'UTR(long)+mtl-2::GFP)]I}, CL2122 {dvIs15[(pPD30.38)unc-54(vector)+(pCL26)mtl-2::GFP]} and DR26 [daf-16(m26)I]. All C. elegans and Escherichia coli strains were obtained from the Caenorhabditis Genetics Center (University of Minnesota) and maintained under standard conditions. Synchronization of nematodes was performed using the standard alkaline hypochlorite method. Nematode experiments were performed at 20°C in S medium with E. coli NA22 as food unless otherwise stated.

2.4. Food clearance assay

Food clearance assay was used to determine the range of polysaccharide concentrations as described (Voisine et al., 2007). Approximately 10-15 synchronized L1 nematodes in 10 μl of S medium were added to 90 μl of NA22 bacterial suspension containing a series of concentrations of polysaccharide in 96-well microplates and incubated at 20°C. The absorbance of the culture was measured at 570 nm every day for 7 days using a Fluoroskan Ascent FL microplate reader (Thermo, Waltham, MA, USA).

2.5. Paraquat survival assay

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Paraquat survival assay using C. elegans was performed as described (Pun et al., 2010) in liquid culture with modifications. Synchronized L1 nematodes were incubated in S medium seeded with E. coli NA22 until L4 and then 75 μg/ml of 5-fluoro-2'-deoxyuridine (FUdR; Sigma, St. Louis, USA) was added to prevent self-fertilization. After further incubation for 24 h, the nematodes were transferred into 96-well microplates (10-20 nematodes/well) containing 70 mM paraquat and polysaccharide samples. The live and dead nematodes (~100 for each treatment) were scored microscopically based on their movement every 12 h until all dead.

2.6. Pgst-4::GFP fluorescence assay

The CL2166 (Pgst-4::GFP) nematodes were treated with or without polysaccharide for 48 h from L1 larvae and then incubated with 10 mM paraquat for 24 h. Fluorescence measurements were then taken on the early-adult nematodes using a COPAS Biosort instrument (Union Biometrica, Inc., Holliston, MA, USA), with lengths measured as time of flight (Choe et al., 2009). For each group, ~300 nematodes were sorted and analyzed for the total green fluorescence and then averaged to one nematode. The Pgst-4::GFP images were captured with an ImageXpress Micro System (Molecular Devices, Sunnyvale, CA, USA).

2.7. Determination of ROS levels

The ROS levels were determined using the 2',7'-dichlorofluorescein diacetate (DCFH-DA; Sigma, St. Louis, USA) as previously described (Wu et al., 2006). Synchronized L1 nematodes were treated with or without polysaccharide and 10 mM paraquat as above. Approximately 1,500 nematodes were collected and washed three times with M9 buffer, and then homogenized in PBST buffer (1×PBS with 0.1% Tween 20) using a glass homogenizer. The lysate was centrifuged at 10,000 g for 5 min at 4°C. Protein concentration was determined with BCA protein assay kit (Thermo, Waltham, MA, USA). A total of 50 μl of

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nematode lysate containing 20 μg proteins was transferred to a 96-well black microplate and incubated with 50 μl of 100 μM DCFH-DA, which is a membrane permeable substance that can enter cells and convert to DCFH. This nonfluorescent probe can be intracellularly oxidized by ROS to produce the fluorescent 2,7-dichlorofluorescein (DCF). Fluorescence (485 nm excitation and 535 nm emission) was monitored on a Fluoroskan Ascent FL microplate reader (Thermo, Waltham, MA, USA) at 25°C every 10 min for 2 h, at which point the data were compared. The changes of fluorescence indicate the variation of ROS levels.

2.8. Determination of malonaldehyde content and superoxide dismutase activity

Determination of malonaldehyde (MDA) content and superoxide dismutase (SOD) activity were performed as previously described (Xiao et al., 2014). Approximately 2,000 synchronized L1 nematodes were incubated with or without polysaccharide and 10 mM paraquat as above. The nematodes were collected and homogenized in 150 μl of Western and IP lysis buffer (Beyotime, Nanjing, China). The lysate was collected and used for protein quantification using BCA kit as above, for SOD activity assay using Total Superoxide Dismutase Assay Kit (Beyotime, Nanjing, China), and for MDA content determination using Lipid Peroxidation MDA Assay Kit (Beyotime, Nanjing, China). The MDA content and SOD activity were normalized by protein content.

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2.9. Mitochondrial membrane potential assay

The mitochondrial membrane potential was determined as previously described (Guo et al., 2014) with modifications. Approximately 800 synchronized nematodes were treated with or without polysaccharide and 10 mM paraquat as above. Freshly collected nematodes were lysed with PBST buffer and protein content was determined as above. A total of 50 μl of nematode lysate containing 20 μg of proteins was transferred to a 96-well black microplate

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and mixed with 50 μl of JC-1 (5,5′,6,6′-tetrachloro-1,1′,3,3′-tetraethylbenzimidazolyl-carbocyanine iodide) working solution according to the manual of the Mitochondrial Membrane Potential Detection kit (Beyotime, Nanjing, China). JC-1 dye, a mitochondrial membrane potential indicator, tends to form J-aggregates in healthy cells with high membrane potential but remains in the monomeric form in apoptotic cells with low membrane potential. The microplate was read at 490 nm excitation and 530 nm emission for JC-1 monomers (green fluorescence) and at 525 nm excitation and 590 nm emission for J-aggregates (red fluorescence) using a Fluoroskan Ascent FL microplate reader (Thermo, Waltham, MA, USA). The ratio of red to green fluorescence was used to monitor the mitochondrial membrane potential.

2.10. ATP measurement

Synchronized nematodes were treated with or without polysaccharide and 10 mM paraquat as above. ATP measurement was performed using an ATP Assay Kit (Beyotime, Nanjing, China) as described (Dong et al., 2013). The nematodes were lysed with the lysis buffer according to the manual of the kit. First, 100 μl of ATP detection solution was added to a white 96-well microplate and incubated for 3-5 min at room temperature, and then 20 μl of the lysate was added. The chemiluminescence detection of ATP was performed using a Fluoroskan Ascent FL microplate reader (Thermo, Waltham, MA, USA). A calibration curve made with ATP standards between 0.01 and 10.0 μM was used to calculate ATP content, which was normalized by protein content.

2.11. Nuclear localization of SKN-1 and DAF-16

SKN-1::GFP nuclear localization was conducted using C. elegans LG345 as described (Chew et al., 2015). Synchronized LG345 nematodes were treated with or without the

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polysaccharide at 20°C from L1 to L2 larvae, in which the SKN-1::GFP expression was reported to be the highest (An and Blackwell, 2003), and then collected for image capturing and scoring of SKN-1::GFP localization. DAF-16::GFP nuclear localization was conducted using C. elegans GR1352 as described (Evason et al., 2008). Synchronized GR1352 nematodes were treated with or without the polysaccharide at 20°C from L1 to L3 larvae, and then collected for image capturing and scoring of DAF-16::GFP localization. For both nuclear localization experiments, the collected samples (~200 nematodes) were transferred to 24-well microplates with ~20 μl of M9 buffer and covered with coverslips. For each group, images of 50 nematodes were acquired with an ImageXpress Micro System (Molecular Devices, Sunnyvale, CA, USA), and the ratio of nematodes with GFP nuclear localization was calculated based on the discrete fluorescent aggregate phenotype in the nematodes.

2.12. Avoidance assay using C. elegans HA759

Avoidance assay with C. elegans HA759 was performed as previously described (Xiao et al., 2014) with modifications. The 2% agar assay plate was divided into normal (N) and trap (T) zones by a glycerol line (8 M, 30 μl) in the middle, and sodium azide (200 mM, 20 μl) was spread about 1 cm away from the glycerol line to paralyze nematodes crossed into the T zone (Fig. 6B). Approximately 300 synchronized N2 (wild type) or HA759 (polyQ) nematodes were treated with or without polysaccharide for 72 h at 20°C until young adulthood, collected and washed with M9 buffer, and then placed on the N zone in four drops. About 2 μl of 1% butanedione was dropped on the T zone (~1 cm from the plate edge) to attract the nematodes. After 1.5 h of incubation at 23°C, the number of nematodes on the two zones was scored as N and T, and the avoidance index was calculated as N/(T+N).

2.13. Chemotaxis assay using C. elegans CL2355

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Chemotaxis assay was performed as previously described (Bargmann et al., 1993) with modifications. Approximately 300 eggs of CL2122 (no Aβ) or CL2355 (Aβ) nematodes were incubated in S medium at 16°C for 36 h until L1 larvae. The nematodes were incubated with or without polysaccharide first at 16°C for 12 h and then at 23°C for 48 h. After collection and three washes with M9 buffer, the nematodes were then transferred to the center of the 2% agar assay plate which was divided into normal (N) and trap (T) zones (Fig. 6C). On the N zone, 1 μl of 100% ethanol (control odorant) and 1 μl of 1 M sodium azide were added to a spot about 1 cm away from the plate edge. On the T zone, 1 μl of 0.1% benzaldehyde (attractant) in ethanol and 1 μl of 1 M sodium azide were added to an opposite spot about 1 cm away from the plate edge. The assay plate was then incubated at 23°C for 1 h and the number of nematodes on the two zones was scored as N and T. Chemotaxis index was calculated as (T-N)/(T+N), where N represented half of the chemosensorily disabled, randomly crawled nematodes with the other half counted in T.

2.14. Statistical analysis

Statistical analysis was performed with GraphPad Prism 5 for windows (GraphPad Software, San Diego, CA, USA). Statistical significance was determined by Student's t test or one-way ANOVA followed by Tukey's post hoc test. Survival analyses were performed using the Kaplan-Meier method and curves were compared for significance by the log-rank test.

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3. Results

3.1. Increase of oxidative survival and decrease of oxidative level in paraquat-intoxicated C. elegans by D. indusiata polysaccharide

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The yield of acidic polysaccharide from Dictyophora indusiata fruit body (DiPS) was 2.2% after ethanol purification and anion-exchange chromatography separation. The total sugar content of DiPS was 80.1%. The molar composition of monosaccharides in DiPS was 86.8% mannose, 4.5% fucose, 3.9% glucose, 1.6% galactose, 1.2% rhamnose, 1.1% glucuronic acid and 0.9% xylose. To evaluate the in vivo antioxidant capacity of DiPS, the survival rate of DiPS-treated C. elegans was monitored after exposure to high doses of paraquat, a widely used superoxide-inducing agent (Rzezniczak et al., 2011). The initial concentration range of the polysaccharide was estimated by food clearance assay (Voisine et al., 2007), which showed concentrations of <4.0 mg/ml were applicable in the wild-type N2 nematodes (Fig. 1A). To test the antioxidant activity and to determine the effective concentrations of the polysaccharide, the nematodes were co-incubated with 70 mM paraquat and 0.0-2.0 mg/ml of DiPS and the survival rates were compared. As shown in Fig. 1B, the survival of paraquat-intoxicated nematodes was improved when treated with 1.0 and 2.0 mg/ml of DiPS, suggesting a protective effect of the polysaccharide against oxidative stress induced by paraquat.

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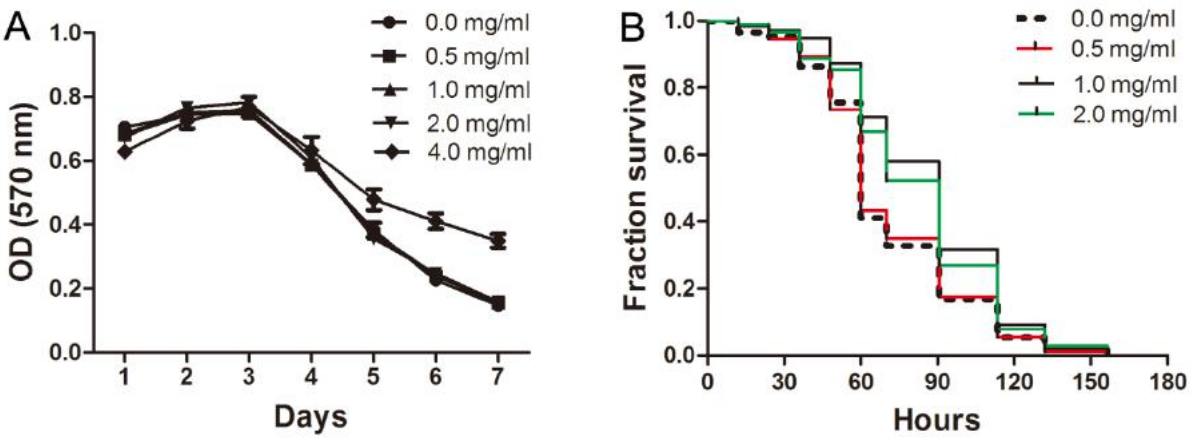

Fig. 1 Effect of Dictyophora indusiata polysaccharide on the survival rate of paraquat-treated Caenorhabditis elegans. (A) The initial concentration range of D. indusiata polysaccharide (DiPS) was estimated by food clearance assay in wild-

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type C. elegans. Synchronized L1 nematodes were incubated at 20°C in 96-well plates with DiPS at the indicated concentrations, and the absorbance at 570 nm was measured daily for 7 days. Results are representative of three independent experiments, and data are shown as mean ± SD of ten parallel wells with 100-150 nematodes per concentration. (B) Oxidative survival curves are shown for wild-type C. elegans treated with 70 mM paraquat and DiPS at the indicated concentrations. The live/dead nematodes (~100 for each treatment) were scored every 12 h until all dead, with hour 0 corresponding to the time of adding DiPS at the beginning of adulthood. Results are representative of three independent experiments, and data are presented as Kaplan-Meier curves and compared for significance by the log-rank test.

To further investigate the antioxidant activity of DiPS, we examined its capability to counteract oxidative stress using the transgenic C. elegans strain CL2166, which carries a GFP reporter expressed from the gst-4 promoter (Leiers et al., 2003). As the expression of C. elegans gene gst-4 is inducible by oxidative stress, the intensity of GFP fluorescence can be used to assess the status of oxidative stress in this nematode model (Wen et al., 2012). As shown in Fig. 2, the polysaccharide itself did not induce Pgst-4::GFP expression but the nematodes treated with 10 mM paraquat displayed an increased GFP level. When the nematodes were co-treated with 10 mM paraquat and 1.0 mg/ml of DiPS, the GFP level was significantly lower than that in the nematodes treated with paraquat alone, demonstrating the ability of the polysaccharide to decrease the oxidative stress level in this nematode model.

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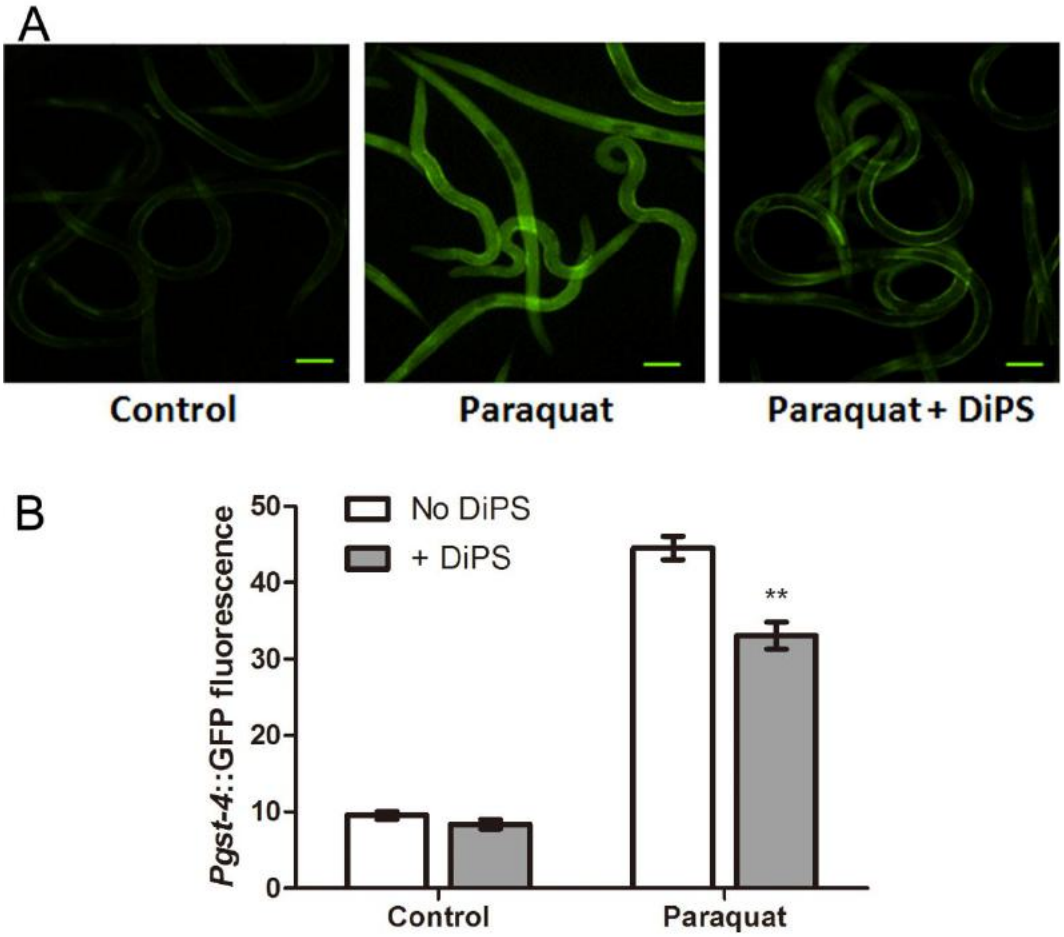

Fig. 2 Effect of Dictyophora indusiata polysaccharide on oxidative stress status in Caenorhabditis elegans. (A) Fluorescence images of Pgst-4::GFP were taken from young adults of C. elegans CL2166. Prior to image capture, the nematodes were treated with or without 1.0 mg/ml of DiPS from L1 for 48 h and then further incubated with 10 mM paraquat for 24 h. Scale bars, 20 μm. (B) Fluorescence intensity of Pgst-4::GFP was determined using the young adult nematodes as in (A). Approximately 300 nematodes for each treatment were used and averaged to one nematode. Results are presented as mean ± SEM of three independent experiments, and statistical significance is determined by Student's t test. ** $P < 0.01$.

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3.2. Decrease of ROS and MDA levels and increase of SOD activity in paraquat-stressed C. elegans by D. indusiata polysaccharide

Since paraquat is known to increase ROS generation, we then tested whether the antioxidant activity of DiPS in C. elegans was through modulation of ROS level. The wild-type nematodes were treated with 10 mM paraquat and 1.0 mg/ml of polysaccharide, and the ROS levels were determined by DCF method. As shown in Fig. 3A, the intensity of DCF fluorescence, which represents relative in vivo ROS level, was considerably increased after paraquat treatment. When the nematodes were co-treated with paraquat and DiPS, the increase of DCF fluorescence was significantly reduced, indicating that the polysaccharide was capable of decreasing elevated ROS level induced by paraquat. As malonaldehyde (MDA), an important biomarker for oxidative stress, is one of the end products of lipid peroxidation triggered by ROS, we further examined the effect of DiPS on MDA content in nematodes treated with 10 mM paraquat. As shown in Fig. 3B, the MDA content was increased by paraquat itself but treatment with 1.0 mg/ml of DiPS reduced the increase, suggesting that the polysaccharide was able to reduce lipid peroxidation. In addition, since superoxide dismutase (SOD) is a major ROS-scavenging enzyme in the endogenous antioxidant defense system, we investigated the effect of DiPS on SOD activity in the nematodes. As shown in Fig. 3C, the SOD activity was increased in the nematodes treated with 10 mM paraquat as compared to the control. Interestingly, the SOD activity was further increased significantly when the nematodes were co-treated with 10 mM paraquat and 1.0 mg/ml of DiPS, indicating the ability of the polysaccharide to increase SOD activity. Together, these results suggest that DiPS can promote the elimination of excessive ROS to alleviate oxidative stress induced by paraquat.

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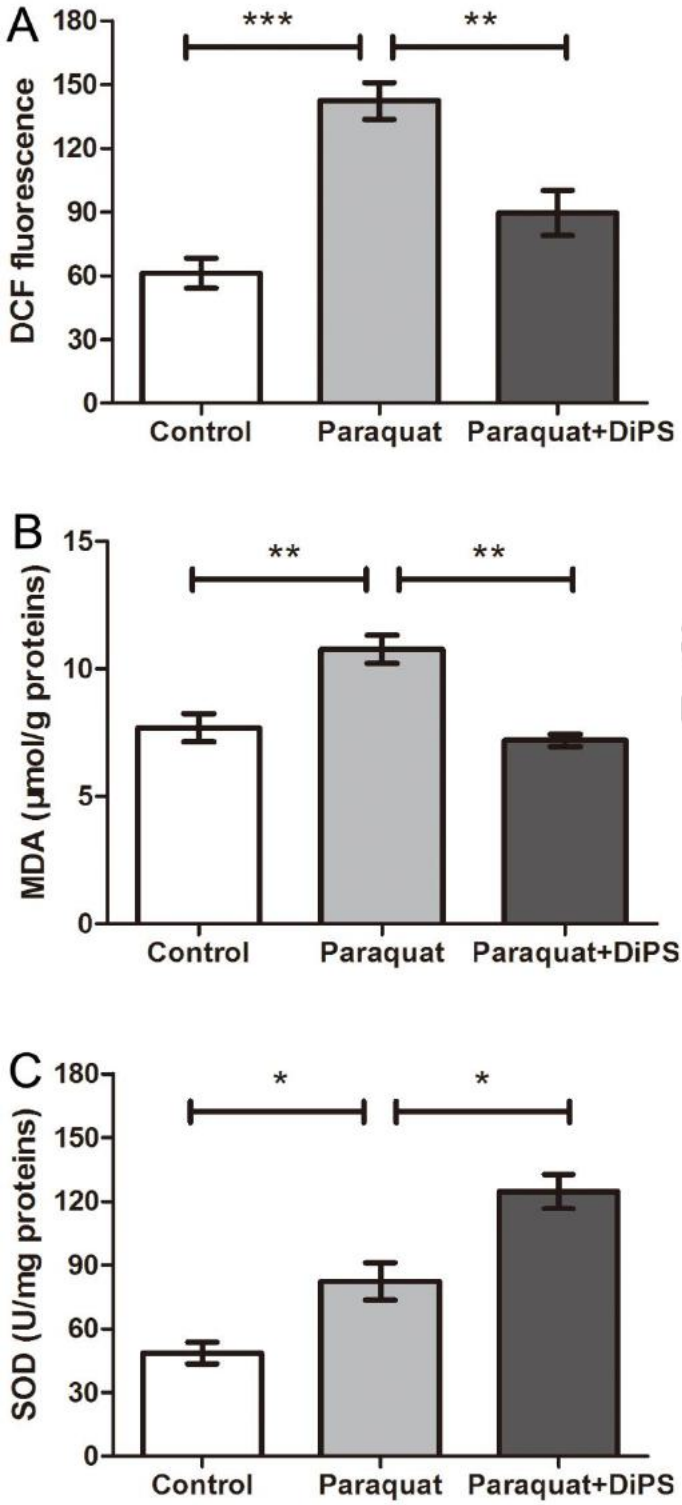

Fig. 3 Effect of Dictyophora indusiata polysaccharide on reactive oxygen species level, malonaldehyde content and superoxide dismutase activity in paraquat-treated Caenorhabditis elegans. Wild-type nematodes were treated as in Fig. 2 before lysis. Relative level of reactive oxygen species (ROS) (A) was determined and compared

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by using equal amount of lysate, while malonaldehyde (MDA) content (B) and superoxide dismutase (SOD) activity (C) were normalized to protein content of the lysate. Data are presented as mean ± SEM of three independent experiments, and statistical significance is determined by one-way ANOVA followed by Tukey's post hoc test. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

3.3. Restoration of mitochondrial membrane potential and ATP content in paraquat-stressed C. elegans by D. indusiata polysaccharide

Mitochondria are considered as the primary source and at the same time the major target of ROS, and excessive ROS can cause the loss of mitochondrial membrane potential by activating mitochondrial permeability transition (Suski et al., 2012). As demonstrated above, DiPS was able to attenuate paraquat-increased ROS level, therefore we investigated the effect of the polysaccharide on mitochondrial membrane potential. The wild-type nematodes were treated with 10 mM paraquat and 1.0 mg/ml polysaccharide, and the mitochondrial membrane potential was determined by the JC-1 method, in which the fluorescence ratio of different JC-1 forms was used to indicate the level of mitochondrial membrane potential. As shown in Fig. 4A, the treatment with paraquat alone reduced the ratio of red to green fluorescence of JC-1, indicating a decreased mitochondrial membrane potential in the nematodes. When the nematodes were co-treated with 10 mM paraquat and 1.0 mg/ml of DiPS, the ratio of red to green fluorescence was restored to normal levels, indicating that DiPS was able to rescue mitochondrial membrane depolarization in paraquat-intoxicated nematodes. Since the collapse of mitochondrial membrane potential is often correlated with interruption of mitochondrial ATP synthesis, we further examined the effect of DiPS on ATP content. Again, paraquat treatment reduced the ATP content, while co-incubation with 1.0 mg/ml of DiPS significantly elevated ATP content in the paraquat-treated nematodes (Fig.

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4B). Together, these results suggest that the maintenance of mitochondrial function, as indicated by membrane potential and ATP content, may contribute to the antioxidant effect of DiPS.

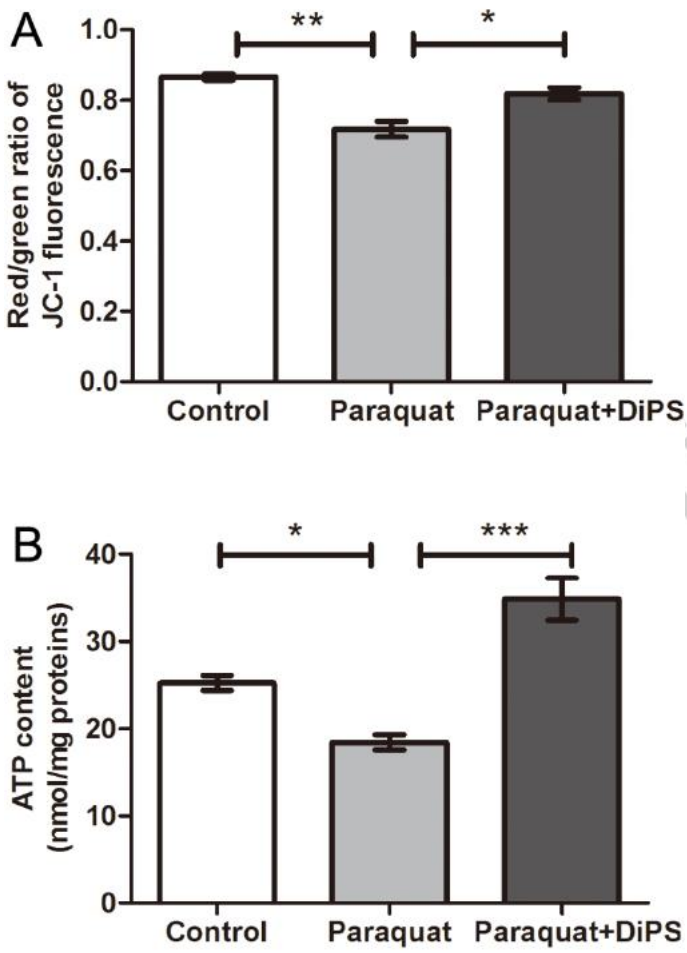

Fig. 4 Effect of Dictyophora indusiata polysaccharide on mitochondrial membrane potential and ATP content in paraquat-treated Caenorhabditis elegans. Wild-type nematodes were treated as in Fig. 2 before used for determination of mitochondrial function parameters. The mitochondrial membrane potential was determined by the JC-1 fluorescence assay and shown as the ratio of red to green fluorescence (A), while the ATP content was determined by chemiluminescence method and normalized to protein content (B). Data are presented as mean ± SEM of three independent experiments, and statistical significance is determined by one-way

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ANOVA followed by Tukey's post hoc test. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

3.4. Involvement of DAF-16 transcription factor in the antioxidant activity of D. indusiata polysaccharide in C. elegans

It is known that ROS act as messenger molecules as well as inducers of oxidative damage in cellular signaling cascades (Ray et al., 2012; Reczek and Chandel, 2015; Schieber and Chandel, 2014). Interestingly, the transcription factors SKN-1/Nrf2 and DAF-16/FOXO, which are associated with stress response and lifespan regulation, have been shown to respond to oxidative stress messengers and activate downstream genes that help increase stress resistance (An and Blackwell, 2003; Zarse et al., 2012). Therefore, we investigated whether DiPS can activate the transcription factors SKN-1 and DAF-16 using nuclear localization assays. The C. elegans strain LG345 expresses skn-1b::GFP fusion proteins in ASI neurons and intestinal cells, and SKN-1::GFP is constitutively present in ASI nuclei and diffusely in intestinal cytoplasm under normal conditions (An and Blackwell, 2003). Since the activation of the SKN-1 signaling pathway requires a translocation of the transcription factor into the nucleus, SKN-1 is regarded as inactive if diffuse fluorescence is present in the nematodes and active if the SKN-1::GFP proteins rapidly accumulates in intestinal nuclei (Chew et al., 2015; Havermann et al., 2014). Therefore, we used this transgenic nematode model to investigate the effect of DiPS on SKN-1 activation but found that the polysaccharide treatment did not alter the percentage of nematodes with detectable nuclear localization (data not shown).

The C. elegans strain GR1352, which has been previously used in the nuclear localization experiments of the DAF-16 forkhead transcription factor (Evason et al., 2008), expresses DAF-16::GFP proteins in many tissues. Upon activation, the fusion proteins can be

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translocated into nucleus from cytoplasm, leading to a discrete fluorescent aggregate phenotype in the nematodes (Havermann et al., 2014). Thus, we used this model to test whether DiPS is able to promote DAF-16 nuclear translocation. As shown in Fig. 5A, treatment with 1.0 mg/ml of DiPS significantly increased the percentage of nematodes with detectable nuclear localization, indicating an activation of DAF-16 by the polysaccharide. To test whether DAF-16 is required for the antioxidant activity of DiPS, we then used C. elegans strain DR26, a daf-16 mutant, to examine the effect of the polysaccharide on the survival and ROS level under oxidative stress. The DR26 nematodes were treated with 70 mM paraquat for survival assay and 10 mM paraquat for ROS determination as above. Both survival rate (Fig. 5B) and ROS level (Fig. 5C) of the daf-16 mutant nematodes remains unchanged after DiPS treatment, demonstrating that the DAF-16 transcription factor is involved in the antioxidant function of the polysaccharide.

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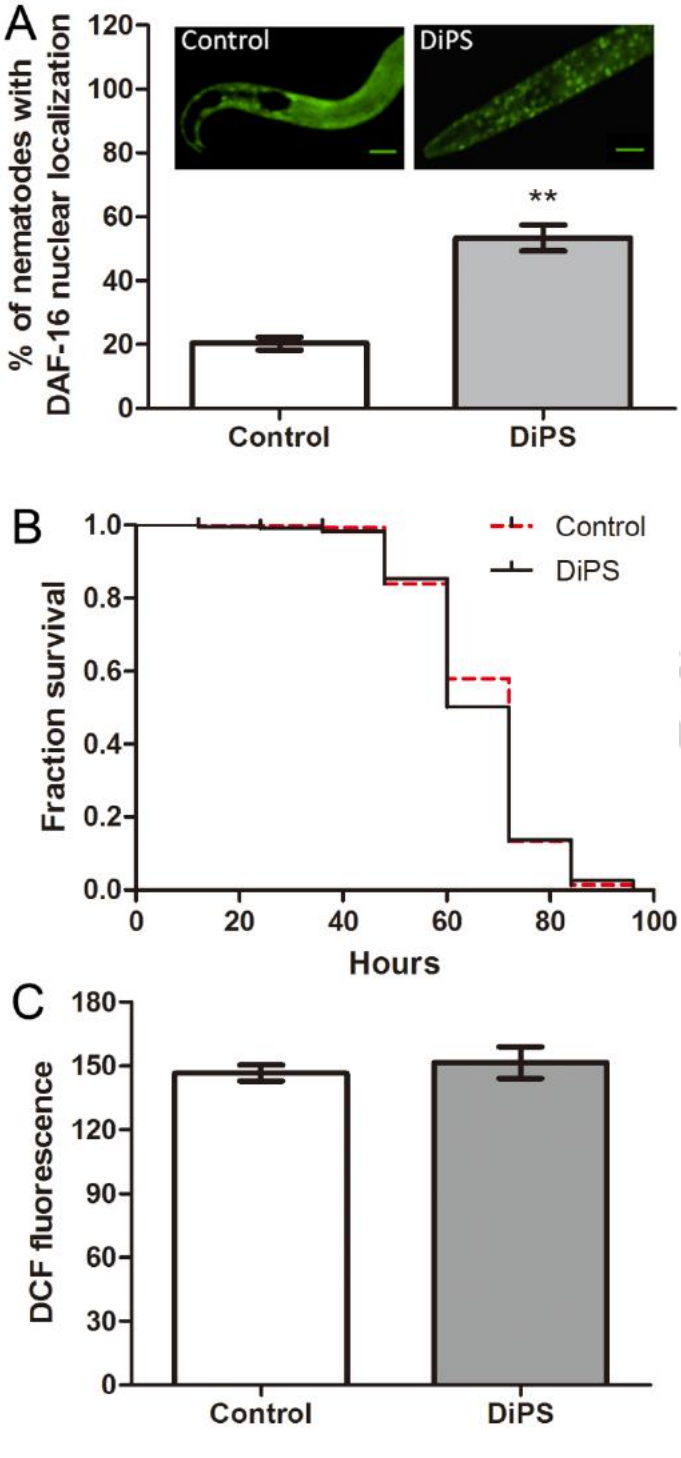

Fig. 5 Effect of Dictyophora indusiata polysaccharide on nuclear translocation of DAF-16 transcription factor and antioxidant activity in daf-16 mutant Caenorhabditis elegans. (A) Percentage of nematodes with DAF-16 nuclear localization was performed using C. elegans GR1352. Synchronized L1 nematodes were treated with or without 1.0 mg/ml of DiPS at 20°C for 48 h before image

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acquisition (scale bars, 100 μm). The ratio of nematodes with DAF-16::GFP nuclear localization was calculated based on the fluorescent puncta, and data are presented as mean ± SEM of three independent experiments. ** $P < 0.01$. (B) Oxidative survival and (C) ROS level of daf-16 mutant C. elegans was performed using the strain DR26 as in Fig. 1B and Fig. 3A, respectively.

3.5. Alleviation of polyQ- and Aβ-mediated chemosensory behavior dysfunction by D. indusiata polysaccharide in C. elegans

Oxidative stress has long been connected to neurodegenerative diseases, e.g. abnormal aggregation of neurodegeneration-related proteins can stimulate ROS generation and cause oxidative damages (Kim et al., 2015). Therefore we speculate the antioxidant polysaccharide DiPS may have the potential to alleviate neurotoxicity induced by protein aggregation in neurodegenerative diseases. To test this, we chose neurodegenerative C. elegans models, including HA759, which expresses polyQ expansions to mimic HD-like phenotypes (Gidalevitz et al., 2006), and CL2355, which expresses Aβ to mimic AD-like phenotypes (Cohen et al., 2006), to examine the neuroprotective capacity of DiPS. Both models show behavioral disorders due to abnormal protein aggregation. We first examined the ROS levels in these models, and found that the treatment with DiPS (1.0 mg/ml) significantly reduced DCF fluorescence in both HA759 and CL2355 nematodes (Fig. 6A), indicating the enhancement of ROS clearance by the polysaccharide in the neurodegenerative models. We then further investigated whether the polysaccharide was able to prevent the behavioral dysfunction of these neurodegenerative C. elegans models using chemosensory assays. The HA759 model expresses polyQ-expanded tracts in ASH neurons, resulting in neurodegeneration and ASH cell death (Faber et al., 1999). Since ASH neurons play important roles in the sensory response of the nematodes to noxious chemical stimuli, we

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used HA759 nematodes to test the effect of DiPS on the avoidance performance to hyperosmotic glycerol. As shown in Fig. 6B, the avoidance index of the wild-type N2 nematodes was >0.9, indicating >90% survival of the ASH neurons. However, the avoidance index of the HA759 nematodes was ~0.5, suggesting that ~50% of the nematodes lost ASH neuronal function. When the HA759 nematodes were treated with 1.0 mg/ml DiPS, the avoidance index was increased to ~0.7, demonstrating that the polysaccharide was able to protect the nematodes from polyQ-mediated neurotoxicity by decreasing the loss of ASH neuronal function. Similarly, we investigated the effect of DiPS on Aβ-mediated neurotoxicity using C. elegans CL2355, which expresses human Aβ1-42 in pan-neurons and causes chemotactic dysfunction (Wu et al., 2006). As shown in Fig. 6C, the chemotaxis indexes of the blank strain CL2122 (no Aβ expression) and the untreated Aβ strain CL2355 were about 0.4 and 0.25, respectively, demonstrating more damage of CL2355 nematodes in neurons sensitive to the attractant benzaldehyde as previously reported (Wu et al., 2006). When the CL2355 nematodes were treated with 1.0 mg/ml DiPS, the chemotaxis index was significantly increased as compared to the untreated CL2355 counterparts, indicating a protective effect of the polysaccharide against Aβ-mediated neurotoxicity by improving chemotactic behavior of the nematodes.

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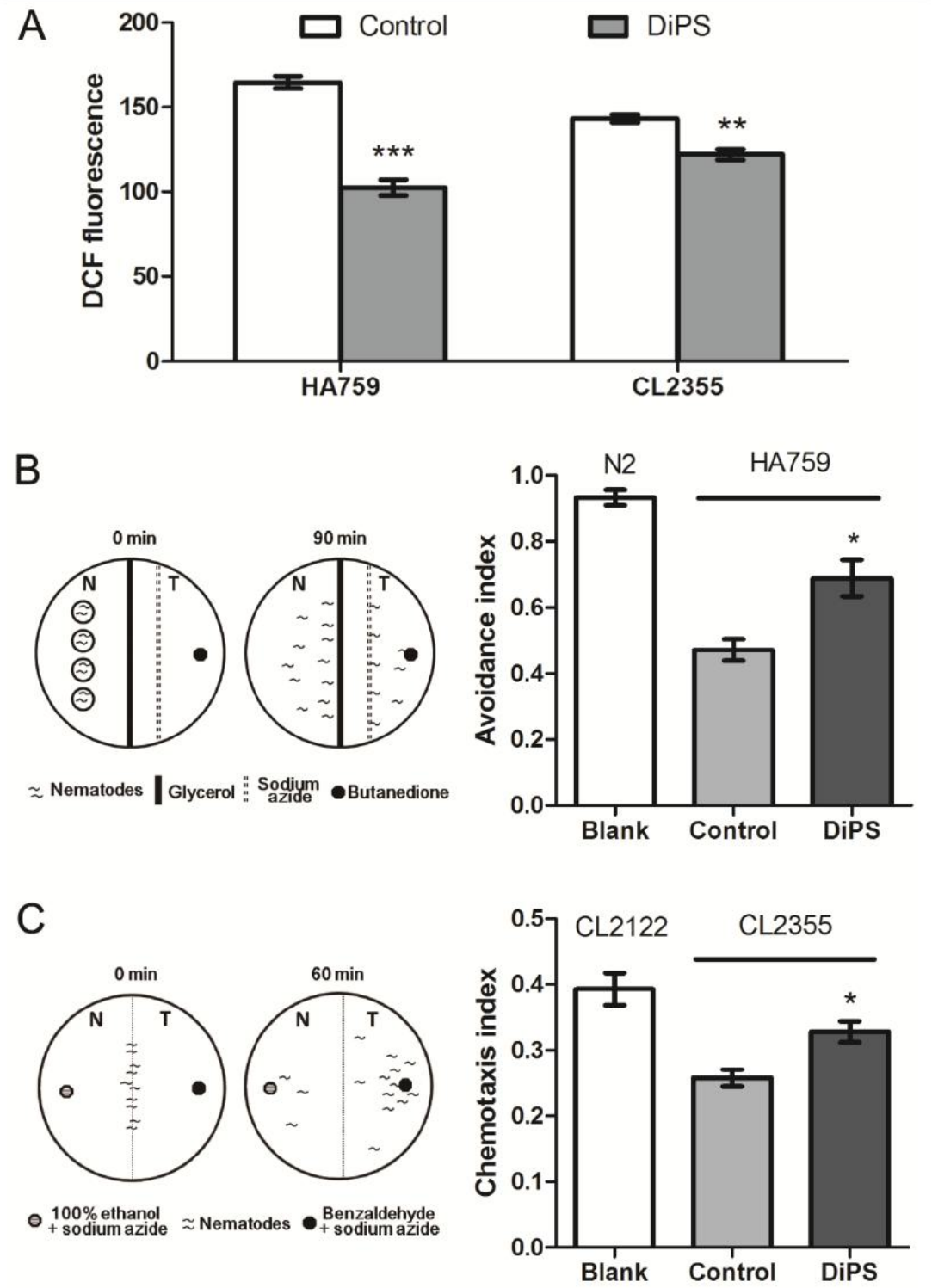

Fig. 6 Effect of Dictyophora indusiata polysaccharide on polyglutamine- and amyloid-β protein-mediated chemosensory behavior dysfunction in Caenorhabditis elegans. (A) ROS levels of polyglutamine (polyQ) and amyloid-β protein (Aβ) nematode models were determined as in Fig. 3A using the strains HA759 and

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CL2355, respectively. (B) PolyQ-mediated behavioral dysfunction was examined by chemosensory avoidance assay in C. elegans HA759. Synchronized nematodes were treated from L1 with or without 1.0 mg/ml of DiPS at 20°C for 72 h and then transferred to the N zone of the assay plate. Glycerol was used as an aversive agent and butanedione as an attractant. Nematodes crawled through the glycerol line were trapped by sodium azide in the T zone, and the nematodes on the two zones were scored respectively. The avoidance index was calculated as the number of nematodes in N zone divided by the total nematodes. (C) Aβ-mediated behavioral dysfunction was examined by chemotaxis assay in C. elegans CL2355. Synchronized nematodes were treated from L1 with or without 1.0 mg/ml of DiPS first at 16°C for 12 h and then at 23°C for 48 h. The nematodes were then transferred to the centre of the assay plate, in which benzaldehyde was used as an attractant in the T zone and ethanol as a control odorant in the N zone. The nematodes on the two zones were scored respectively, and the chemotaxis index was calculated as the adjusted number (T-N) of nematodes trapped in T zone divided by the total nematodes. In both chemosensory behavior assays, approximately 300 nematodes were used for each treatment, and data are presented as mean ± SEM of three independent experiments. Statistical significance is determined by Student's t test (A) and one-way ANOVA followed by Tukey's post hoc test (B and C). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

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4. Discussion

The implication of oxidative stress in ageing process and pathogenesis of age-related diseases has caused an increasing interest in natural antioxidants. For example, the polysaccharides from the fungus Pleurotus abalonus are recently found to have both

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antioxidant and antitumor effects (Ren et al., 2015). We have previously demonstrated that astragalan, a polysaccharide isolated from A. membranaceus, is capable of reducing polyQ-mediated proteotoxicity and extending the lifespan of C. elegans (Zhang et al., 2012), and recent studies have shown that the polysaccharides from A. membranaceus also have antioxidant activities (Huang et al., 2013). Here we show that DiPS, a polysaccharide isolated from the medicinal mushroom Dictyophora indusiata, is capable of increasing the survival rate of C. elegans under oxidative stress (Fig. 1), indicating the in vivo antioxidant capacity of D. indusiata polysaccharide, which is previously reported to have in vitro antioxidant activity (Deng et al., 2012; Ker et al., 2011). Further examinations reveal that DiPS can not only promote the elimination of excessive ROS (Fig. 3) but also alleviate both polyQ- and Aβ-mediated behavioral dysfunction (Fig. 6), demonstrating the neuroprotective function of the polysaccharide in HD- and AD-like C. elegans models.

It has been widely recognized that oxidative stress caused by excessive ROS is closely associated with neurodegenerative diseases (Gandhi and Abramov, 2012; Xie et al., 2013). Therefore, strategies aiming at scavenging free radicals and reducing ROS production are a promising approach against neurodegeneration. For example, salidroside, a phenol glycoside from Rhodiola rosea, is shown to reduce the intracellular ROS level and protect C. elegans from polyQ-mediated neurodegeneration (Xiao et al., 2014). The polysaccharides obtained from Hedysarum polybotrys are shown to scavenge superoxide radicals in vitro and protect human neuroblastoma cells against Aβ-induced oxidative stress (Wei et al., 2015). In the current study, we observed that the polysaccharide DiPS is able to inhibit lipid peroxidation and increase SOD activity in paraquat stressed wild-type C. elegans (Fig. 3), demonstrating its antioxidant activities. Interestingly, we also found that DiPS is capable of reducing the ROS levels not only in paraquat-treated wild-type nematodes but also in polyQ and Aβ models (Fig. 3 and Fig. 6), indicating the potential of the polysaccharide to alleviate

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neurotoxicity. Further experiments using behavioral assays revealed that DiPS is indeed able to rescue behavioral dysfunctions in HD- and AD-like C. elegans models (Fig. 6). Taken together, these results not only demonstrate the neuroprotective effect of the polysaccharide through ROS regulation but also support a correlation between excessive ROS and neurodegenerative disease mechanisms in general.

Mitochondria are the main source of ROS generation which is caused by electron leakage from the oxidative phosphorylation pathway (Rigoulet et al., 2011). Excessive ROS is likely to cause mitochondrial dysfunction, which in turn will promote ROS production (Facecchia et al., 2011). Therefore, reducing ROS at the source represents an efficient strategy to maintain mitochondrial function and alleviate oxidative damages. For example, a polysaccharide from Ganoderma atrum is recently shown to inhibit overproduction of ROS and prevent mitochondrial dysfunction (Li et al., 2015). Interestingly, it has also been documented that mitochondrial dysfunction is associated with the pathogenesis of neurodegenerative disorders such as AD and HD (Cabezas-Opazo et al., 2015; Hoffmann et al., 2014). Therefore, restoring mitochondrial function is likely an effective approach to alleviate neurodegenerative manifestations. For instance, apigenin, a dietary flavonoid found in plants, has been reported to attenuate copper-mediated β-amyloid neurotoxicity through its mitochondria-protecting and antioxidant activities (Zhao et al., 2013). A recent study has also shown that the polysaccharide from Polygonatum sibiricum is able to inhibit mitochondrial dysfunction and prevent Aβ-induced neurotoxicity and cell death (Zhang et al., 2015). In this study, we found that the polysaccharide DiPS can restore paraquat-induced reduction of mitochondrial membrane potential and ATP content, which are important functional parameters of mitochondria, in wild-type nematodes (Fig. 4), indicating the potential of the polysaccharide to maintain mitochondrial function under conditions of oxidative stress. In addition, we also demonstrate that DiPS is able to improve the chemosensory behavior in

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transgenic polyQ and Aβ nematode models, suggesting the neuroprotective capacity of the polysaccharide through ameliorating disrupted mitochondrial function. Interestingly, we have recently shown that the polysaccharide from Angelica sinensis has the potential to facilitate the recovery of rats from cerebral ischemia and reperfusion injury, and demonstrated that the protective effect of the polysaccharide against nerve cell injury caused by oxidative stress is associated with its antioxidant activities, including prevention of mitochondrial membrane potential decrease (Lei et al., 2014). Taken together, these results provide an insight into the potential of antioxidant polysaccharides as neuroprotective agents.

The DAF-16/FOXO transcription factor is a critical mediator against oxidative stress (Furukawa-Hibi et al., 2005), and thus regulation of DAF-16/FOXO pathway is a viable approach to reduce oxidative damages. For example, resveratrol, a well-recognized antioxidant, has been shown to increase mitochondrial function and protect cells from oxidative stress by activating SIRT1-FOXO pathway (Hori et al., 2013; Yun et al., 2012). Caffeic acid phenethylester, an active constituent of propolis, is recently shown to have a strong antioxidant effect and extend lifespan in C. elegans by modulation of the DAF-16 signaling pathway (Havermann et al., 2014). In the present study, we found that DiPS is able to promote the translocation of DAF-16 into the nucleus in C. elegans models (Fig. 5), demonstrating the activation of the DAF-16 pathway by the polysaccharide. Importantly, DiPS is capable of increasing the survival rate and reducing the ROS level in wild-type (Fig. 1 and 3), but not daf-16 mutant (Fig. 5), C. elegans under oxidative stress induced by paraquat, indicating a protective effect of the polysaccharide against oxidative stress in a DAF-16-dependent manner. Interestingly, DAF-16 is also reported to act as a critical modulator of proteotoxicity induced by Aβ aggregation (Cohen et al., 2006), and we have previously found that astragalan, the polysaccharide from A. membranaceus, is able to protect polyQ-induced proteotoxicity in a DAF-16-dependent fashion (Zhang et al., 2012).

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Therefore, we tested the neuroprotective potential of DiPS in HD- and AD-like C. elegans models, and found that DiPS is indeed able to reduce both polyQ- and Aβ-mediated chemosensory dysfunction (Fig. 6). Since DAF-16 transcription factor plays a pivotal role in a variety of signaling pathways related to both stress resistance and lifespan regulation (Hsu et al., 2003; Putker et al., 2013), we also examined the effect of DiPS on the lifespan of nematodes but no lifespan extension was found (data not shown). Intriguingly, however, the neuroprotective polysaccharide astragalan is able to extend the lifespan of the nematodes in a DAF-16-dependent manner (Zhang et al., 2012). Collectively, these results suggest that different polysaccharides may exert their activities through distinct but also overlapping pathways depending on specific context.

5. Conclusion

In this study, we demonstrate the in vivo antioxidant capacity of D. indusiata polysaccharide (DiPS) through an increased oxidative survival and a reduced oxidative status in paraquat-intoxicated C. elegans models. Then we reveal that DiPS is capable of reducing ROS and MDA levels and increasing SOD activity, as well as restoring mitochondrial function in terms of membrane potential and ATP content, in paraquat-stressed nematodes. We also show that DAF-16/FOXO, a key transcription factor related to both stress response and lifespan regulation, is involved in the antioxidant activity of the polysaccharide. Further experiments using HD- and AD-like nematode models demonstrate that DiPS can not only reduce ROS level in neurodegenerative models but also improve polyQ- and Aβ-mediated chemosensory behavior disorders, indicating the neuroprotective capacity of the polysaccharide. These results not only demonstrate the neuroprotective functions of the traditional mushroom D. indusiata in neurodegeneration but also provide an important insight

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into the potential of polysaccharides from traditional medicinal herbs as antioxidant-based therapeutics in neurodegenerative diseases.

Authors' contributions

Ju Zhang, Haifeng Li and Zebo Huang conceived and designed the research. Ju Zhang, Ruona Shi, Haifeng Li, Yanxia Xiang and Lingyun Xiao performed the experiments. Ju Zhang, Haifeng Li, Minghua Hu, Fangli Ma and Chung Wah Ma analyzed the data. Ju Zhang, Haifeng Li and Zebo Huang wrote and revised the manuscript. All authors have read and approved the final version.

Conflict of interest

The authors disclose no conflicts of interest.

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